# Day 15 | Wednesday

15

November 25, 2026

## Reading and substituting into formulas

12:45 PM–2:00 PM • Math • 75 min

## Rearranging formulas and checking backwards

2:15 PM–3:30 PM • Math • 75 min

**Week 5 • 2 scheduled blocks • 150 teaching minutes**

Estimate. Show the method. Label units. Check the result.

# Today’s exact schedule

## 12:45 PM–2:00 PM • 75 min

### Reading and substituting into formulas

| | | |
|---|---|---|
| 12:45 PM–12:50 PM | **Start** | 5 |
| 12:50 PM–12:55 PM | **Recall** | 5 |
| 12:55 PM–1:10 PM | **Teach** | 15 |
| 1:10 PM–1:25 PM | **Together** | 15 |
| 1:25 PM–1:45 PM | **Practise** | 20 |
| 1:45 PM–1:55 PM | **Check** | 10 |
| 1:55 PM–2:00 PM | **Wrap up** | 5 |

Math • Math Week 5, Lesson 3

## 2:15 PM–3:30 PM • 75 min

### Rearranging formulas and checking backwards

| | | |
|---|---|---|
| 2:15 PM–2:20 PM | **Start** | 5 |
| 2:20 PM–2:25 PM | **Recall** | 5 |
| 2:25 PM–2:40 PM | **Teach** | 15 |
| 2:40 PM–2:55 PM | **Together** | 15 |
| 2:55 PM–3:15 PM | **Practise** | 20 |
| 3:15 PM–3:25 PM | **Check** | 10 |
| 3:25 PM–3:30 PM | **Wrap up** | 5 |

Math • Math Week 5, Lesson 4

**Break: 2:00 PM–2:15 PM • 15 minutes • excluded from teaching time**

# Reading and substituting into formulas

**Today’s goal: Name variables and substitute dimensions into a formula with correct brackets and units.**

Describe a prior use of the skill; identify the given and unknown quantities.

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

Exercise Book 8A Q1–8 (p.77); 8B Q12 (p.80)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

# Recall and reconnect

Identify what l and w stand for in A = l × w. Check BEDMAS using a two-step expression.

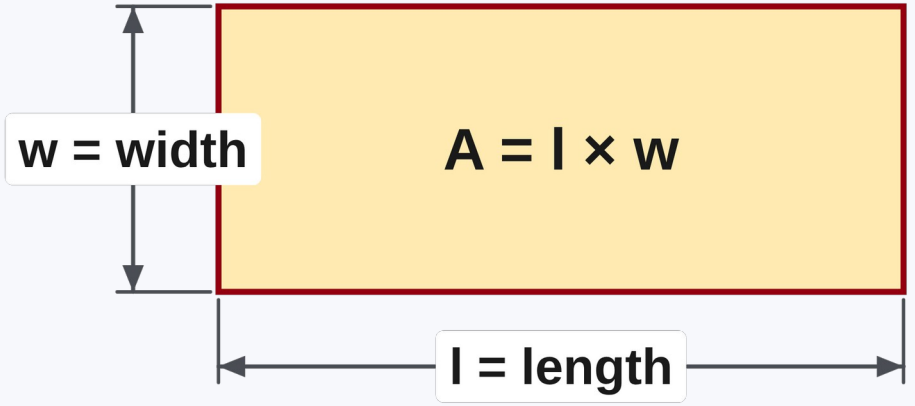
**Attempt the retrieval task independently; explain the method before checking.**

Resources: Exercise Book 8A Q1–8 (p.77); 8B Q12 (p.80)

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

# Identify what each symbol means

1. List the given measurements and the quantity you need to find.
2. Match each formula letter to the correct labelled dimension.
3. Use consistent units before substituting values.
4. Keep brackets where the formula groups operations.




Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

# Follow a formula from symbols to a result

1. Write the formula before entering the numbers.
2. Substitute the known values in the same positions.
3. Calculate in the correct order and attach the resulting unit.
4. Check both the magnitude and meaning: equal area alone does not prove a part fits.

**Formula** written first → **Substitute** known values → **Calculate** in order → **Unit and** check

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

# Match the formula to the requested quantity

Length l = 4 m



**Width w = 2 m**

**A = l × w**
**= 4 × 2 = 8 m²**

P = 2(l + w)
= 2(4 + 2) = 12 m

Together: 8A Q2–4 and Q7. Name each quantity before keying.
Numeracy Hub Shape Studio: compare letters, values and calculations.

W5 L3 • Classroom rectangle, not a scale drawing • Area and perimeter are different quantities

# Textbook | page 193

## Example 8.1 Area and perimeter of a plate

A plate is 1,200 mm long and 800 mm wide. Use $A = l \times w$ to find its area and $P = 2 \times (l + w)$ to find the distance around it.

**SOLUTION**

1. Area: $A = 1{,}200 \times 800 = 960{,}000$.
2. Perimeter: $P = 2 \times (1{,}200 + 800)$. Brackets first, $1{,}200 + 800 = 2{,}000$. Then $2 \times 2{,}000 = 4{,}000$.

Textbook p.193 • Example 8.1 — Substitution into area and perimeter formulas

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

# Textbook | page 193

**CALCULATOR** Perimeter

**TI-30Xa**

2 × ( 1200 + 800 ) =

Display 4000.

**Sharp EL-531XT**

2 × ( 1200 + 800 ) =

Display 4’000.

Key the brackets. Without them the calculator works 2 × 1,200 first and then adds 800.

**Answer** **A = 960,000 mm² (square millimetres) and P = 4,000 mm.**

Textbook p.193 • Example 8.1 — Brackets, calculator checks and units

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

# Spot the mistake. Explain the correction.

## If two rectangles have the same area, must one fit exactly over the other?

Discuss the method before checking the answer.

**No. Compare their length and width as well; equal products can have different dimensions.**

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

# Work together

Work Exercise Book 8A Q2–4 and Q7. Name each quantity before entering calculator keys.

**Work in pairs, alternating solver and checker. Write a diagram, formula, units and estimate.**

Resources: Exercise Book 8A Q1–8 (p.77); 8B Q12 (p.80)

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

# Guided practice | 8A Q2

Rewrite 3ab and πr² with × signs.

Work with a partner. Take turns explaining the next step.

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

# Compare your method | 8A Q2

Rewrite 3ab and πr² with × signs.

**CHECK TOGETHER**

**3 × a × b and π × r × r**

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

# Your scheduled practice

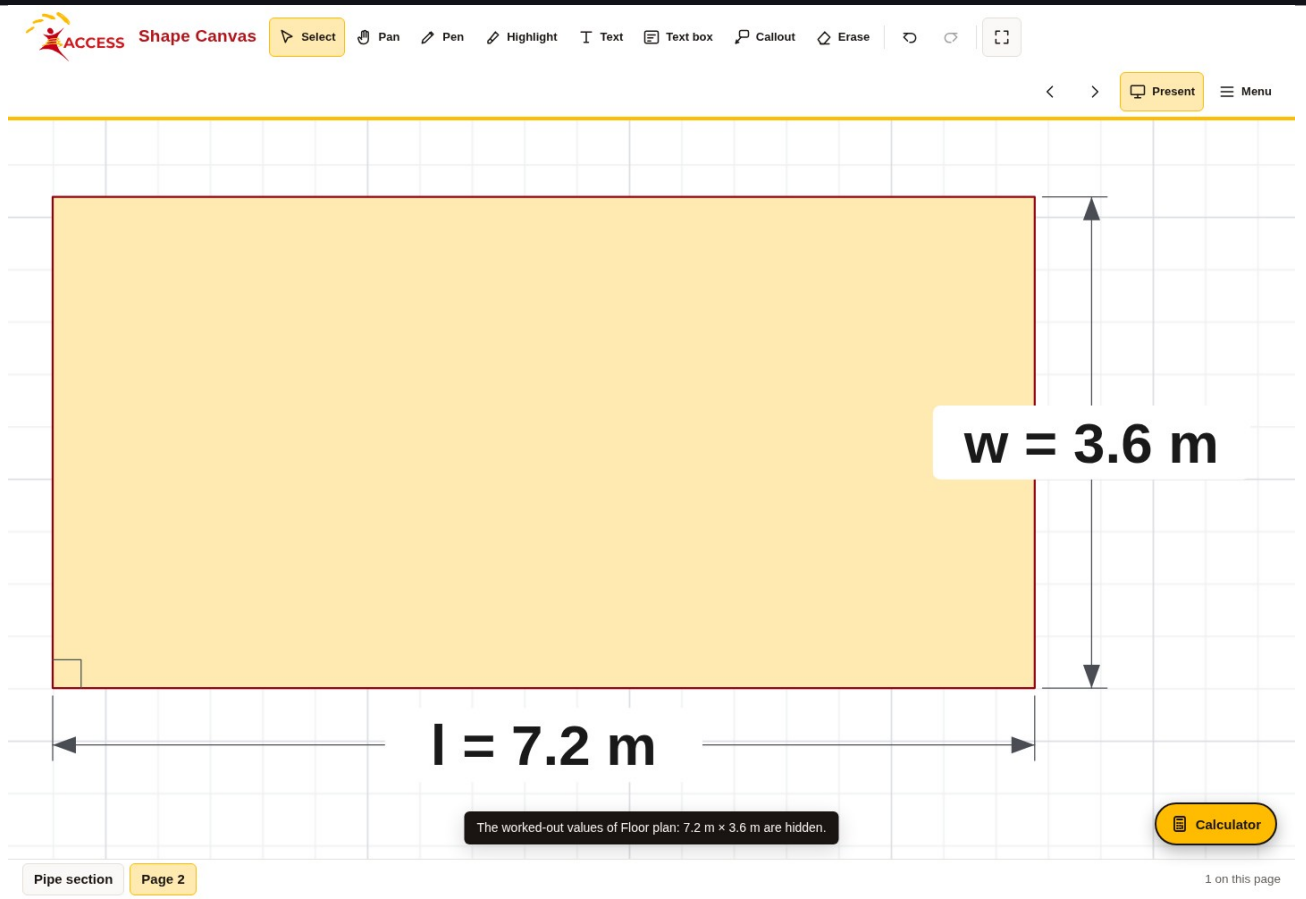
Complete Exercise Book 8A Q5–6,8 and 8B Q12. Use Shape Canvas dimensions to verify substituted values; treat formula applications as introductions, not full geometry mastery.

**Solve on paper first; check with tools; correct one error and explain the change.**

Resources: Exercise Book 8A Q1–8 (p.77); 8B Q12 (p.80)

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

# Open Shape Canvas




Navigation example. Use the values and units in the assigned task.

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

# Practise in Numeracy Hub and Shape Canvas

1. In Numeracy Hub Shape Studio, compare Letters and values with Calculations.

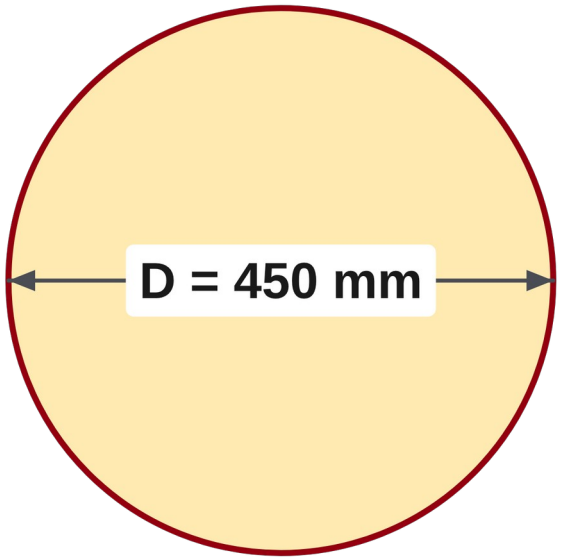
2. Create the assigned rectangle in Shape Canvas and show its sizes and formula.

3. Substitute the labelled dimensions on paper, then compare with the tool calculation.

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

# Practise from the Exercise Book | 8A Q5

Substitute: $C = \pi \times D$ with $D = 450$ mm. Round to two decimal places.



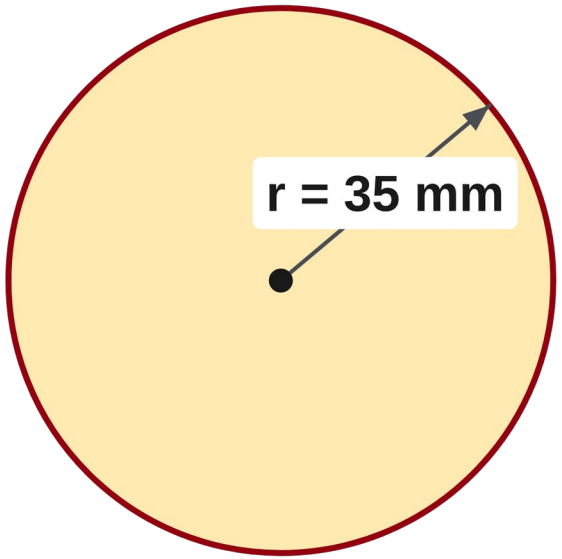

Circumference C ?

Write the operation, working and units before checking.

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

# Practise from the Exercise Book | 8A Q6

Substitute: $A = \pi \times r^2$ with $r = 35$ mm. Round to two decimal places.




Area A ?

Write the operation, working and units before checking.

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

# Check independently

Exercise Book 8A Q3–4; label the result as area or perimeter and include units.

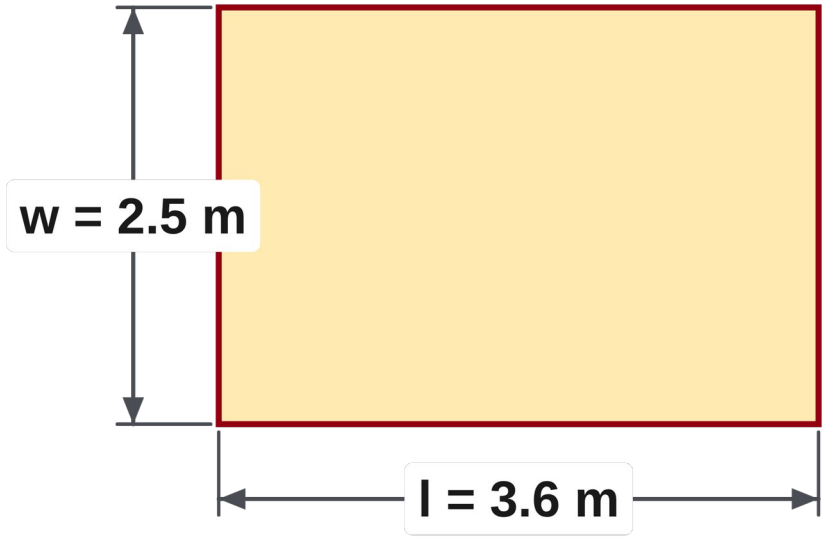
**Solve without displayed answers; include working, units and final rounding.**

Resources: Exercise Book 8A Q1–8 (p.77); 8B Q12 (p.80)

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

# Independent check | 8A Q3

Substitute: $A = l \times w$ with $l = 3.6$ m and $w = 2.5$ m.



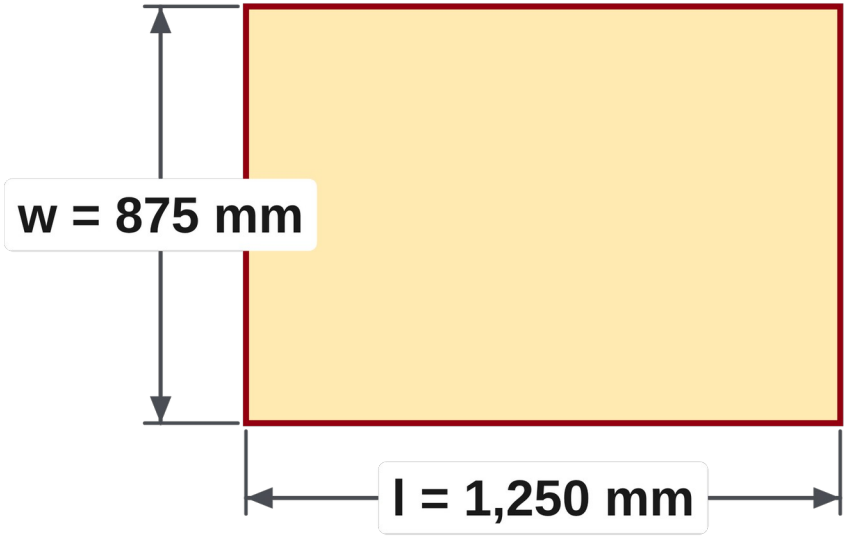

Area A ?

Answer independently. Keep worked solutions closed.

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

# Independent check | 8A Q4

Substitute: P = 2 × (l + w) with l = 1,250 mm and w = 875 mm.




Perimeter P ?

Answer independently. Keep worked solutions closed.

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

# Wrap up and choose the next step

Ask: What changed in your understanding? Where will you use it? What should you check next time?

**Name the math and one other essential skill used; record a next practice step.**

Resources: Exercise Book 8A Q1–8 (p.77); 8B Q12 (p.80)

Textbook sections 8.1–8.2 pp.192–194; formula sheet p.361

# Break

## 2:00 PM–2:15 PM

Return ready for: Rearranging formulas and checking backwards

**This break is excluded from teaching minutes.**

# Rearranging formulas and checking backwards

**Today’s goal: Rearrange one- and two-step formulas and validate a missing dimension by substitution.**

Describe a prior use of the skill; identify the given and unknown quantities.

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

Exercise Book 8A Q9–22 (pp.77–78), selected; 8B Q1,4 (p.79), Q15 (p.80)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

# Recall and reconnect

Use Exercise Book 8A Q9,11. Discuss which inverse operation preserves equality.

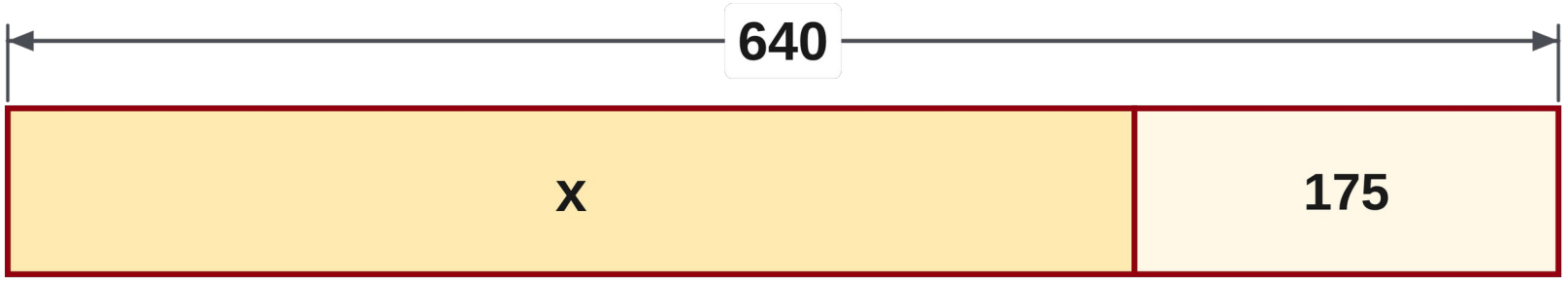
**Attempt the retrieval task independently; explain the method before checking.**

Resources: Exercise Book 8A Q9–22 (pp.77–78), selected; 8B Q1,4 (p.79), Q15 (p.80)

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

# Recall from the Exercise Book | 8A Q9

Solve x + 175 = 640.




Write the operation, working and units before checking.

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

# Undo operations on both sides

1. Identify the unknown before rearranging the formula.
2. Use inverse operations: add/subtract and multiply/divide.
3. Apply the same operation to both sides to preserve equality.
4. Keep the entire grouped expression together when dividing.

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

# Check a missing dimension backwards

1. For P = 2(l + w), divide by 2 before subtracting the known width.
2. The rearranged formula is l = P/2 − w.
3. Insert the found length into the original perimeter equation.
4. Accept the result only when the original equation balances with correct units.

**P = 2(l + w)**
start

→

**P ÷ 2 = l + w**
divide by 2

→

**l = P ÷ 2 − w**
subtract w

→

**Check**
l back in P

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

# Balance the equation, then check

$$P = 2(l + w) \rightarrow P \div 2 = l + w \rightarrow l = P \div 2 - w$$

**Worked example • P = 4,000 mm; w = 700 mm.**
**l = 4,000 ÷ 2 − 700 = 1,300 mm.**
**Check: 2 × (1,300 + 700) = 4,000 mm.**

Spot the error • l = P − 2w gives 2,600 mm—not the length.
Substituting that answer gives 6,600 mm, not the stated perimeter.

Together • Exercise Book 8A Q15,18–19 and 8B Q1.
Say the inverse operation; keep both sides equal. Check every result in the original formula before using the calculator or Shape Canvas.

Textbook pp.194–200,203–204 • Exercise Book pp.77–80 • Worked example: Exercise Book 8B Q15

# Textbook | page 199

## Example 8.8 Find the length from the perimeter

A rectangular frame uses 5,000 mm of flat bar all the way around. It is 900 mm wide. Transpose $P = 2 \times (l + w)$ to find the length.

**SOLUTION**

1. Undo the × 2: divide both sides by 2. $P \div 2 = l + w$
2. Undo the + w: subtract w from both sides. $l = P \div 2 - w$
3. $l = 5{,}000 \div 2 - 900 = 2{,}500 - 900 = 1{,}600$

Textbook p.199 • Example 8.8 — Undo two operations to find a length

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

# Textbook | page 199

**CALCULATOR**

**TI-30Xa**

5000 ÷ 2 − 900 =

Display 1600.

**Sharp EL-531XT**

5000 ÷ 2 − 900 =

Display 1′600.

**Answer** 1,600 mm. Check: 2 × (1,600 + 900) = 5,000.

Textbook p.199 • Example 8.8 — Substitute back into the original perimeter formula

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

# Spot the mistake. Explain the correction.

## Why does l = P − 2w give the wrong length for a rectangle?

Discuss the method before checking the answer.

**After subtracting 2w, the left side is still 2l. Divide the remaining expression by 2.**

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

# Work together

Complete Exercise Book 8A Q15,18–19 and Exercise Book 8B Q1 as a group. Explain the inverse step.

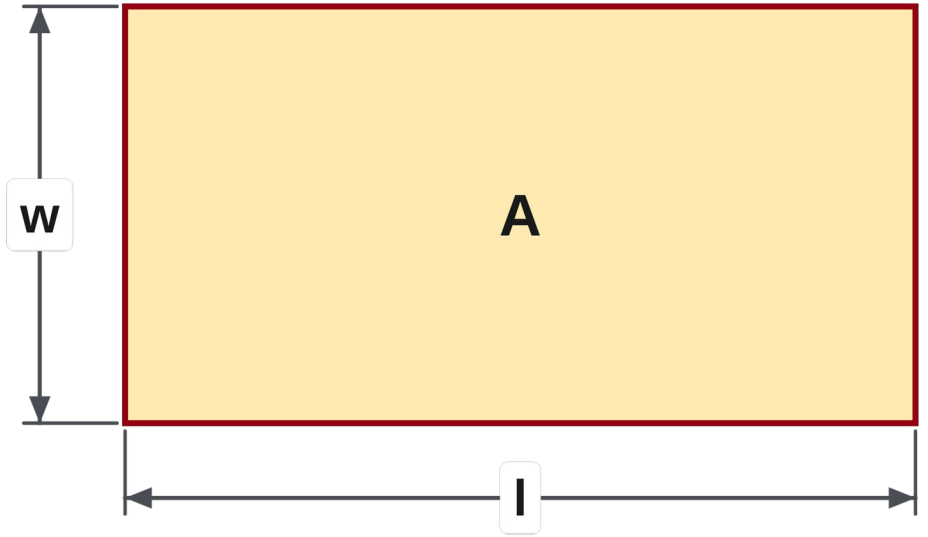
**Work in pairs, alternating solver and checker. Write a diagram, formula, units and estimate.**

Resources: Exercise Book 8A Q9–22 (pp.77–78), selected; 8B Q1,4 (p.79), Q15 (p.80)

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

# Guided practice | 8A Q15

Make l the subject of A = l × w.




Work with a partner. Take turns explaining the next step.

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

# Compare your method | 8A Q15

Make l the subject of A = l × w.

**CHECK TOGETHER**

**l = A ÷ w**

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

# Your scheduled practice

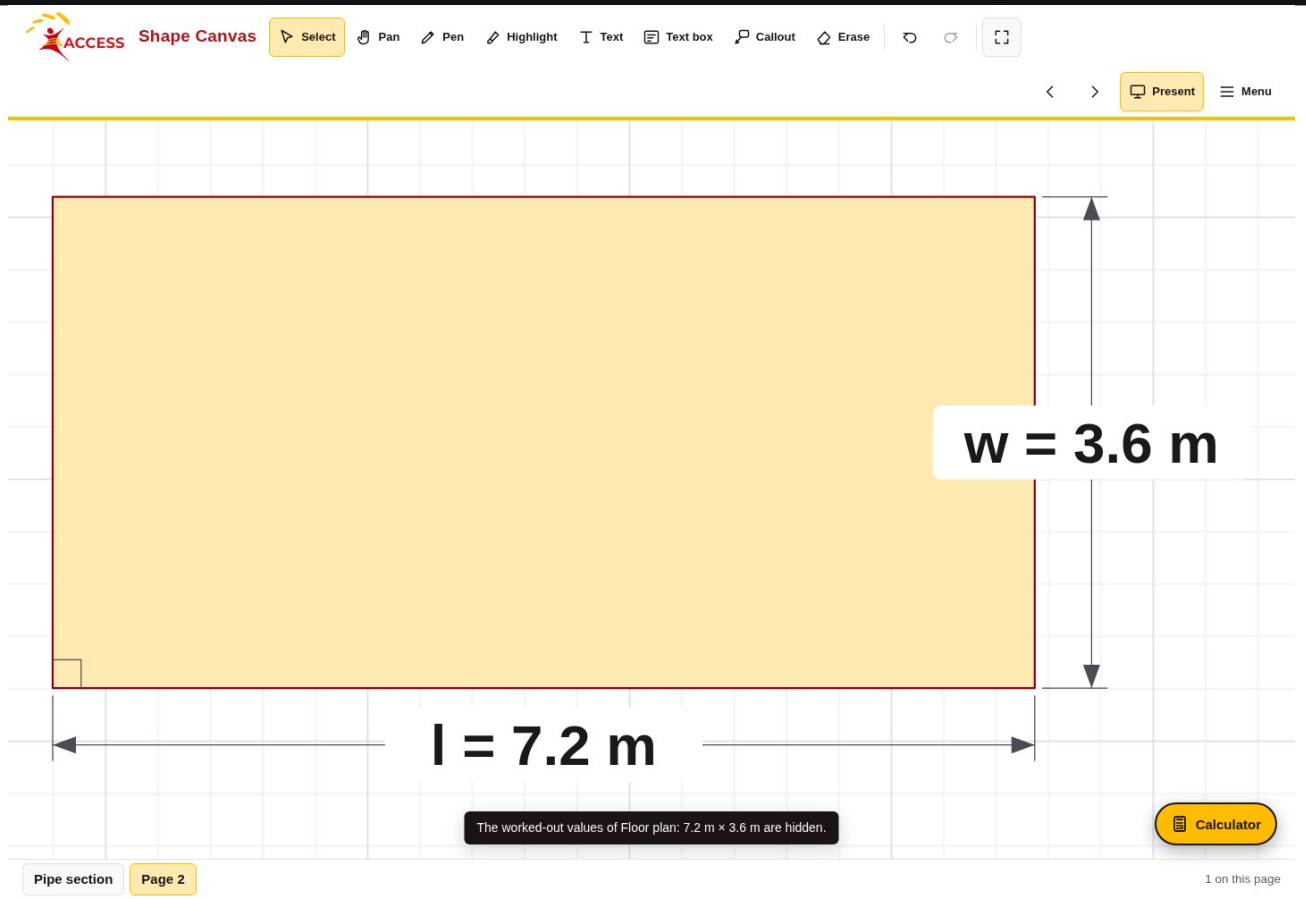
Solve selected Exercise Book 8A Q12–14,16–17,20–22 and Exercise Book 8B Q4,15. Check every result in the original equation; use tools after the written rearrangement.

**Solve on paper first; check with tools; correct one error and explain the change.**

Resources: Exercise Book 8A Q9–22 (pp.77–78), selected; 8B Q1,4 (p.79), Q15 (p.80)

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

# Open Shape Canvas




Navigation example. Use the values and units in the assigned task.

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

# Practise in Numeracy Hub and Shape Canvas

1. Use a Shape Canvas rectangle with known area or perimeter and one unknown dimension.

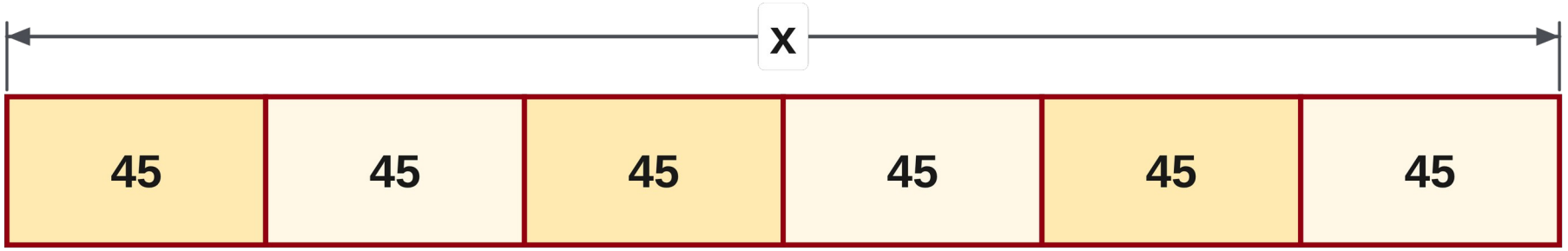
2. Rearrange the original formula and solve the missing dimension on paper.

3. Use Numeracy Hub Calculator to substitute back into the original formula and check equality.

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

# Practise from the Exercise Book | 8A Q12

Solve x ÷ 6 = 45.

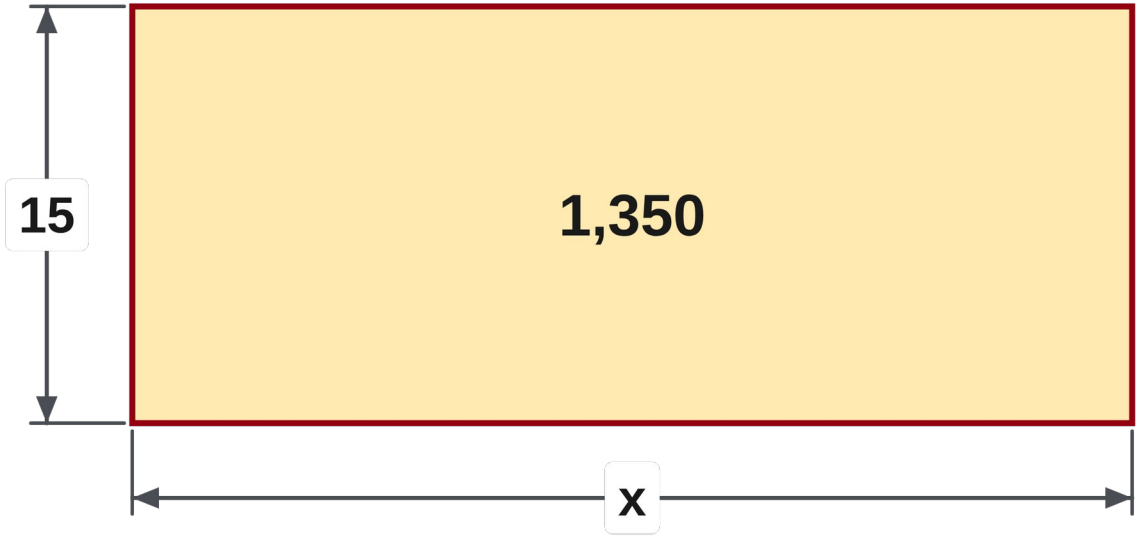



Write the operation, working and units before checking.

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

# Practise from the Exercise Book | 8A Q13

Solve 15 × x = 1,350.




Write the operation, working and units before checking.

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

# Check independently

Exercise Book 8B Q4 and Q15; show the substitution check, not just the answer.

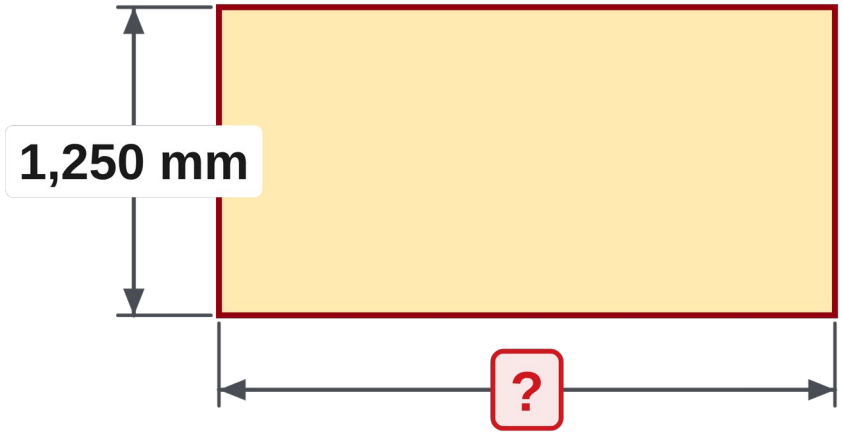
**Solve without displayed answers; include working, units and final rounding.**

Resources: Exercise Book 8A Q9–22 (pp.77–78), selected; 8B Q1,4 (p.79), Q15 (p.80)

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

# Independent check | 8B Q4

A welded frame uses 6,400 mm of tube all the way around. It is 1,250 mm wide. How long is it?




All the way around: 6,400 mm

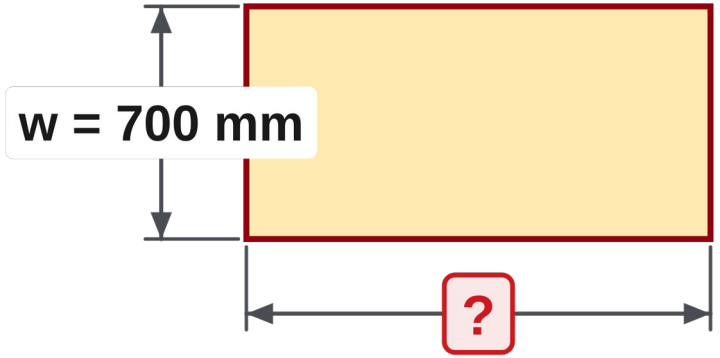
Length ?

Answer independently. Keep worked solutions closed.

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

# Independent check | 8B Q15

Find the mistake. A learner transposes $P = 2 \times (l + w)$ to $l = P - 2 \times w$ and uses $P = 4{,}000$ mm and $w = 700$ mm. Find the learner's answer, check it by substituting back, and find the correct length.




P = 4,000 mm

Length l ?

Answer independently. Keep worked solutions closed.

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

# Wrap up and choose the next step

Ask: What changed in your understanding? Where will you use it? What should you check next time?

**Name the math and one other essential skill used; record a next practice step.**

Resources: Exercise Book 8A Q9–22 (pp.77–78), selected; 8B Q1,4 (p.79), Q15 (p.80)

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204